Lecture 2

Repeated Games

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Repeated Games

- Prisoner's Dilemma (PD) suggests cooperation is hard to sustain with self-interested players
- "Shadow of future interactions" might help sustain cooperative outcomes (after all, they occur in reality)
- Sometimes we want to study the same basic interaction repeated multiple times
- To analyze **repeated games**, we need a bit of extra technology

Discounted Payoffs

In a repeated game, players receive payoffs every period

Denote by $u_i(x^t)$ player i's payoff from outcome x^t in period t

Suppose a sequence of action profiles yields a stream of payoffs $(u_i(x^1), u_i(x^2), \ldots)$

Payoffs tomorrow are typically not as valuable as payoffs today (forgone interest, unforeseen events, etc.)

Players **discount** the future at rate $\delta \in (0, 1)$

Discounted Payoffs

δ is also known as “time preference,” measures one’s patience

- If δ close to 1, care a lot about future
- If δ close to 0, it’s pure living in the moment (*carpe diem*)
- Plausibility of assumption about δ depends on how a time interval is defined in the specific application
- E.g., re-election in six years vs negotiating another bill tomorrow

A Useful Fact

Fact

$$\sum_{t=1}^{T} \delta^{t-1} = \frac{1-\delta^T}{1-\delta} \quad ; \quad \sum_{t=1}^{\infty} \delta^{t-1} = \frac{1}{1-\delta}$$

Why?

$$\begin{aligned} Z &= 1 + \delta + \delta^2 + \ldots + \delta^{T-1} \\ \delta Z &= \phantom{1 + {}} \delta + \delta^2 + \ldots + \delta^{T-1} + \delta^T \end{aligned}$$

Subtracting yields $Z(1-\delta) = 1 - \delta^T \xrightarrow[T\to\infty]{} 1$

Average Discounted Payoffs

A player evaluates a sequence of outcomes $(x^1, x^2, \ldots)$ as

$$V = u_i(x^1) + \delta u_i(x^2) + \delta^2 u_i(x^3) + \ldots = \sum_{t=1}^{\infty} \delta^{t-1} u_i(x^t)$$

For each stream $(u_i(x^1), u_i(x^2), \ldots)$, there is $c \in \mathbb{R}$ such that i is indifferent btw the stream and a constant stream $(c, c, \ldots)$, i.e.,

$$V = \sum_{t=1}^{\infty} \delta^{t-1} u_i(x^t) = \sum_{t=1}^{\infty} \delta^{t-1} c = \frac{c}{1-\delta}$$

Hence, we can write $c = (1-\delta)V$, and call c is the **average discounted payoff** of the stream $(x^1, x^2, x^3, \ldots)$

c is an affine transformation of V: they represent the same preferences

Defining a Repeated Game

Let $G = \{N; A_1, \ldots, A_n; u_1, \ldots, u_n\}$ be a strategic form game with perfect information

Definition

G_T, the T**-period repeated game of** G for the discount factor δ is the extensive form game in which

- The set of players is N
- $\mathcal{H} = \{(a^1, \ldots, a^t) \in A^t\}_{t=1}^{T}$
- The set of terminal histories is the set of sequences $(a^1, a^2, a^3, \ldots, a^T)$ of action profiles in G
- The action space for any player i after any history is A_i
- Each player i evaluates each terminal history according to its discounted payoff: $\pi_i(a^1, \ldots, a^T) = \sum_{t=1}^{T} \delta^{t-1} u_i(a^t)$

Infinitely Repeated Games

The **infinitely repeated game of G (G_∞) for discount factor δ** differs from the T-period repeated game of G only in that the set of terminal histories is the set of infinite sequences $(a^1, a^2, \ldots)$

The finitely repeated prisoner's dilemma (PD)

		Pl. 2	
		C	D
Pl. 1	C	$2, 2$	$0, 3$
	D	$3, 0$	$1, 1$

THE PRISONER'S DILEMMA

- At stage T, both players will *Defect*
- At stage $T - 1$, both players know they will choose *Defect* in final stage, so *Defect*
- Unique SPE is one in which both players *Defect* after every history

Another Finitely Repeated Game

	A	B	C
A	$9, 9$	$1, 6$	$-2, 10$
B	$6, 1$	$5, 5$	$1, 0$
C	$10, -2$	$0, 1$	$2, 2$

Claim: (s', s') is a Nash Equilibrium whenever $\delta \geq 1/3$:

1. If she follows s' player i obtains discounted payoff $9 + 5\delta$
2. Given that player $-i$ plays s', s'' cannot be a profitable deviation for i if it changes second period behavior only
3. suppose $s''(\emptyset) = B$, then i can obtain at most $6 + 2\delta < 9 + 5\delta$
4. suppose $s''(\emptyset) = C$, then i can obtain at most $10 + 2\delta$. Whenever $\delta \geq 1/3$, $10 + 2\delta < 9 + 5\delta$

Exercise

- Is (D, D) in both periods the unique Nash Equilibrium outcome of PD_T?
- Show that the strategy described in the previous slide induces a Nash Equilibrium in every subgame

The infinitely repeated PD (PD_∞)

- In the game above, cooperation is sustained by using multiple equilibria in second period
- Finitely repeated PD unravels to no cooperation because there is only one NE in final period and so on
- What happens if the game does not have a last period so can't unravel? Can we create a logic similar to our other example?
- Can we sustain mutual cooperation (C, C)?

Strategies in PD_∞

A strategy for player i specifies an action following every history of the game $(a^1, a^2, \ldots, a^t)$

We call the following strategy the **grim trigger** :

- $s_i(\emptyset) = C$
- $s_i(a^1, \ldots, a^t) = \begin{cases} C & \text{if } (a^1_{-i}, \ldots, a^t_{-i}) = (C, \ldots, C) \\ D & \text{else} \end{cases}$

for every history $(a^1, \ldots, a^t)$, where $-i$ is the other player

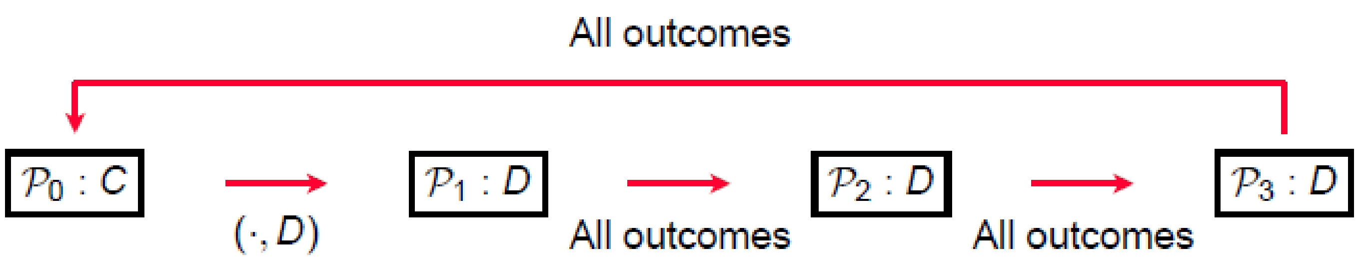
Two "states"

1. Start in **Cooperation**; always choose C in that state
2. Switch to **Punishment** if $-i$ ever chose D, stay there forever

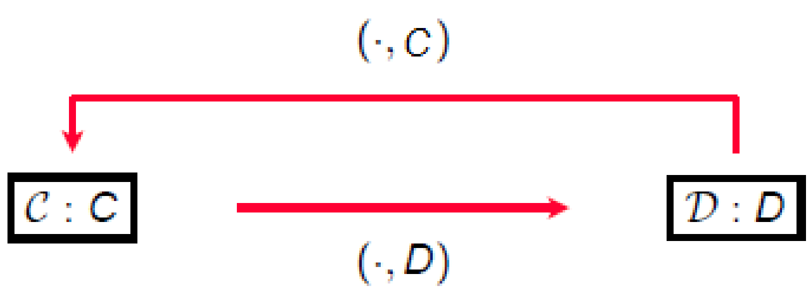
Pictures of strategies in PD_∞

- Grim Trigger

- Three period punishment



- Tit-for-Tat



NE of the PD_∞

The infinitely repeated PD (PD_∞) is an extensive form game: we can apply our definitions to see if a strategy profile is a NE

Specify a strategy profile, then ask whether any player can make herself better off by unilaterally choosing a different strategy

Does PD_∞ admit a NE in which players do not always *Defect* on one another?

Is (Grim Trigger,Grim Trigger) a NE?

Generic strategy s', which prescribes its first D at period t

- Payoff stream from s is at most: $(2,\ldots,2,3,1,1,1,\ldots)$

$$ADP: (1-\delta)\Big(2(1+\delta+\ldots+\delta^{t-2})+3\delta^{t-1}+\delta^{t}+\delta^{t+1}+\ldots\Big)$$
$$=(1-\delta)2\sum_{j=1}^{t-1}\delta^{j-1}+(1-\delta)3\delta^{t-1}+(1-\delta)\delta^{t-1}\delta\sum_{j=1}^{\infty}\delta^{j-1}$$

Suppose i follows Grim Trigger

- Payoff stream from following strategy: $(2,2,2,2,\ldots)$

$$ADP: (1-\delta)2(1+\delta+\ldots)$$
$$=(1-\delta)2(1+\delta+\ldots)=(1-\delta)2\sum_{j=1}^{t-1}\delta^{j-1}+(1-\delta)2\delta^{t-1}\sum_{j=1}^{\infty}\delta^{j-1}$$

To compare, cancel the common terms

Is (Grim Trigger,Grim Trigger) a NE?

We obtain

$$(1-\delta)3\delta^{t-1} + (1-\delta)\delta^{t-1}\delta\sum_{j=1}^{\infty}\delta^{j-1} \leq (1-\delta)2\delta^{t-1}\sum_{j=1}^{\infty}\delta^{j-1}$$
$$\Leftrightarrow\ (1-\delta)\delta^{t-1}\left(3+\delta\frac{1}{1-\delta}\right) \leq (1-\delta)2\delta^{t-1}\frac{1}{1-\delta}$$

Deviation is not profitable if and only if (iff)

$$3(1-\delta)+\delta \leq 2 \iff \delta \geq 1/2$$

Is Both Playing a k-Period Punishment a NE?

Suppose Pl. 2 deviates to D in round t

- Pl. 1 chooses D from period $t+1$ to period $t+k$, regardless of Pl. 2's subsequent choices
- Pl. 2's best deviation also chooses D in these periods
- In period $t+1+k$, Pl. 2 is in the same situation as in t
- Profitability must come during periods t to $t+k$:

$$(1-\delta)\left(3\delta^{t-1}+\sum_{i=1}^{k}\delta^{t-1+i}\right)=\delta^{t-1}\big(3(1-\delta)+\delta(1-\delta^{k})\big)$$

Playing proposed profile from round t to $t+k$ yields

$$(1-\delta)(2\delta^{t-1}+2\delta^{t}+2\delta^{t+1}+\ldots+2\delta^{t+k-1})=2\delta^{t-1}(1-\delta^{k+1})$$

Is k-Period Punishment a NE (cont.)?

Profile is a NE if and only if:

$$2\delta^{t-1}(1-\delta^{k+1}) \geq \delta^{t-1}\left(3(1-\delta)+\delta(1-\delta^{k})\right)$$
$$\iff 2\delta - 1 - \delta^{k+1} \geq 0$$

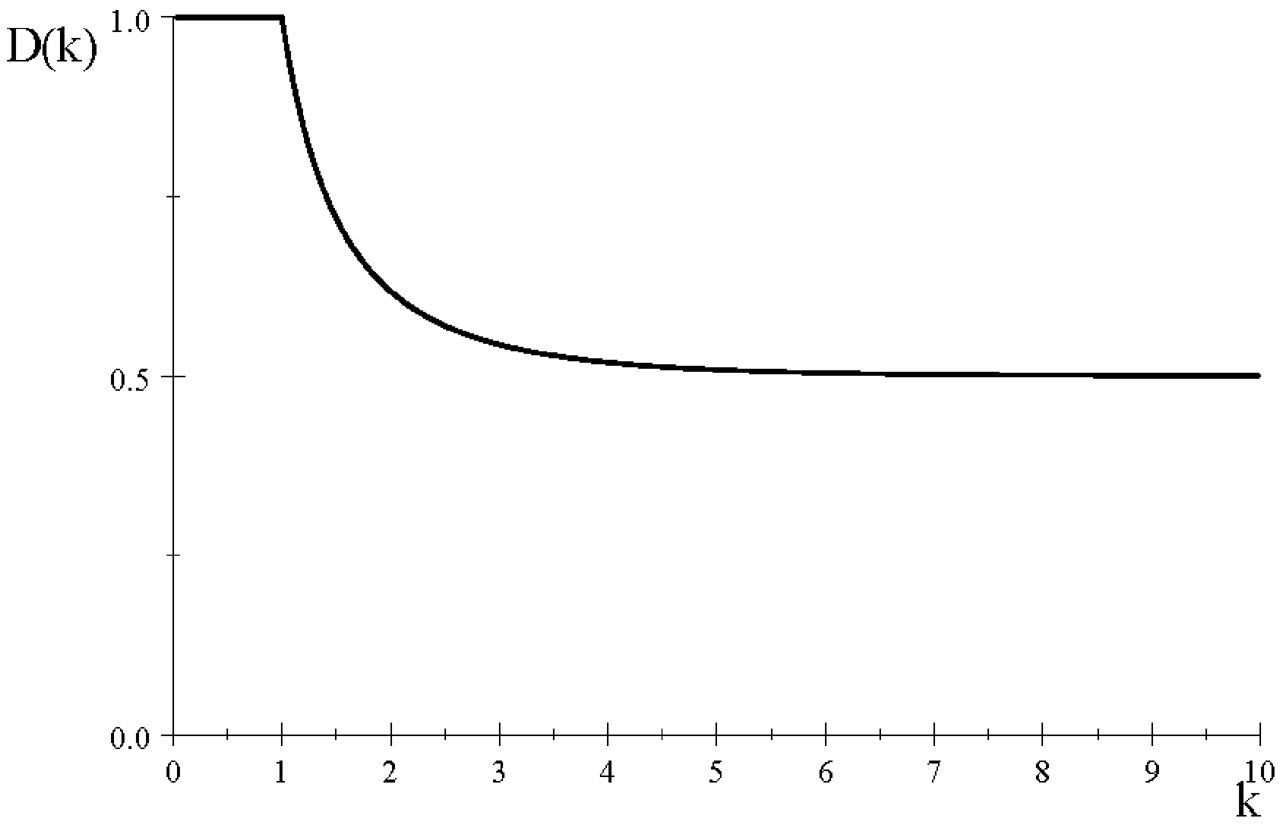
Left-hand side (LHS) is increasing in k and increasing in δ whenever LHS ≤ 0

Able to sustain equilibrium if $\delta \geq D(k)$ for some $D(k)$

$D(k)$ is decreasing in k (the higher k, the lower δ can be)

$D(\infty) = \frac{1}{2}$ and $D(0) = 1$

Graphically



Is Both Playing Tit-for-Tat a NE?

Suppose Pl. 2 chooses D in period t

In period $t+1$, Pl. 1 chooses D

- If Pl. 2 chooses D as well, then D forever: $(3, 1, 1, \ldots)$
- If Pl. 2 chooses C, then alternate: $(3, 0, 3, 0, \ldots)$

Profile is an equilibrium iff

$$2\delta^{t-1} \geq \delta^{t-1}\left(3(1-\delta)+\delta\right) \iff \delta \geq \frac{1}{2}$$

and

$$2\delta^{t-1} \geq \delta^{t-1}\left((1-\delta)\frac{3}{1-\delta^2}\right) \iff \delta \geq \frac{1}{2}$$

Some Things to Notice

PD_∞ admits many equilibria

Some of them, for some parameter values, involve "cooperative" outcomes

But both playing "always choose D" is always an equilibrium

A natural question is then "what payoffs can be sustained in equilibrium?"

Convexifying the Payoff Space

Outcome path in which (X, Y) occurs every period yields average discounted payoffs $(u_1(X, Y), u_2(X, Y))$

In PD_∞, there are outcome paths which yield $(2, 2), (3, 0), (0, 3)$ and $(1, 1)$ as pairs of discounted average payoffs

If outcome alternates between any two of these players, payoffs will be an average of the two average discounted payoffs

If we change frequency of occurrence of each, we get different weights on average pairs of average discounted payoffs

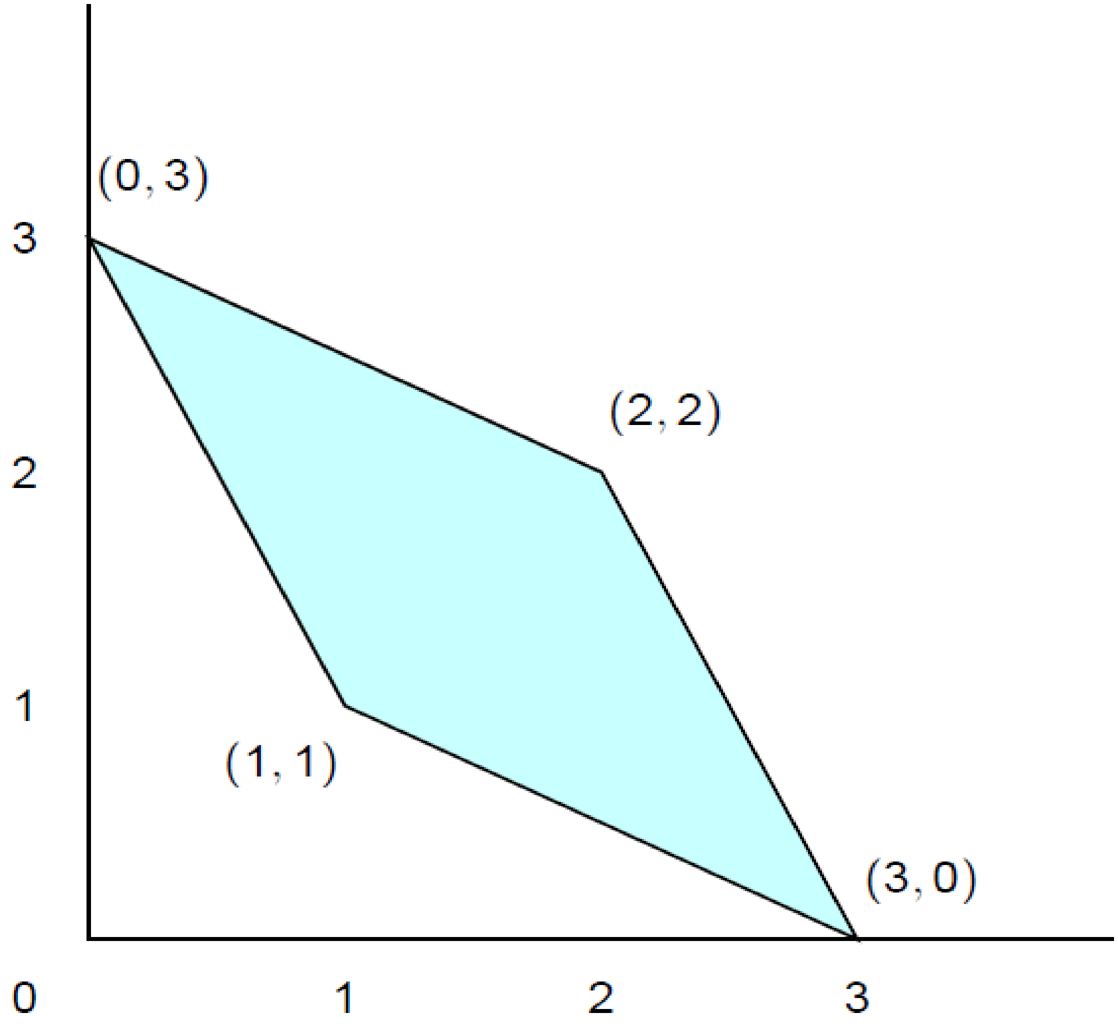
Feasible Payoffs

Definition

The set of **feasible payoff profiles** in game G is the set of all weighted averages of pure strategy payoff profiles of G

For any feasible payoff in the stage game G, there exists a sequence of actions in G_∞ whose average discounted payoffs equal those payoffs

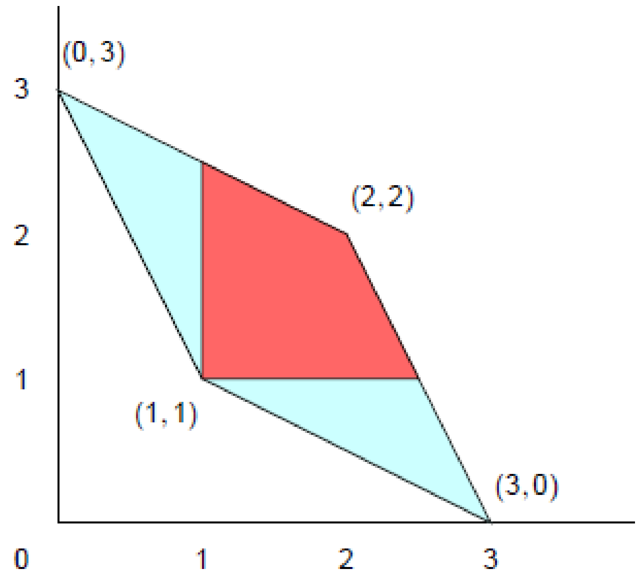
Feasible Average Discounted Payoffs in PD_∞



What Payoffs Can Be Part of a NE?

By choosing D in every round, a player can ensure a payoff of 1

There cannot be a Nash Equilibrium where a player obtains an Average Discounted Payoff of less than 1



THEOREM: All other pairs of payoffs are sustainable in NE, as long as δ is large enough

Sustaining an Arbitrary Feasible Payoff Pair

Suppose we want to sustain feasible payoffs (x_1, x_2) in equilibrium, with $\min(x_1, x_2) > 1$

Let $(b^1, b^2, \ldots)$ be the outcome path that generates (x_1, x_2)

Start playing action called for under $(b^1, b^2, \ldots)$, continue to do so as long as other player does it, if not D forever

- A sufficient condition for this to be a NE is that
$$\min(x_1, x_2) \geq 3(1-\delta) + \delta \iff \delta \geq \frac{3 - \min(x_1, x_2)}{2}$$
- Since $x_1, x_2 > 1$, this is true for some $\delta < 1$

The Nash Folk Theorem for PD_∞

Proposition

Proposition 1.

- *For any discount factor* $\delta \in (0,1)$*, the discounted average payoff of each player* i *in any NE of* PD_∞ *is at least* $u_i(D,D)$
- *Let* (x_1, x_2) *be a feasible pair of payoffs in* PD *for which* $x_i > u_i(D,D)$ *for each* i*; there exists a* $\overline{\delta} < 1$ *such that if* $\delta \geq \overline{\delta}$*, then* PD_∞ *has a NE with discounted average payoff* (x_1, x_2)
- *For any value of the discount factor,* PD_∞ *has a NE in which the average discounted payoff is* $u_i(D,D)$

Subgame Perfection in the PD_∞

From earlier discussion, NE may entail non-credible threats

Subgame Perfection addresses this issue by requiring optimal play at every subgame

We implemented SPE through backward induction

How can we implement the same idea in an infinitely repeated game (where we cannot backward induct)?

The Grim Trigger and SPE

We labeled to the following strategy the Grim Trigger

$$\boxed{\mathcal{C} : C} \xrightarrow[(\cdot, D)]{} \boxed{\mathcal{D} : D}$$

This strategy clearly involves threats (if you *Defect* against me, I will *Defect* against you for ever after)

Are these threats credible for δ sufficiently large?

- Take the subgame following (C, D) in period 1
- If both adhere to Grim Trigger they should play (D, C) in period 2 and (D, D) in period 3 and forever after
- This is clearly not optimal for Pl. 2 in period 2; the strategy profile is a NE, but not a SPE for any δ

Can we find a strategy that is immune to this problem?

The One Shot Deviation Principle

We learned that some strategy profiles that are NE in the infinitely repeated game may not be SPE

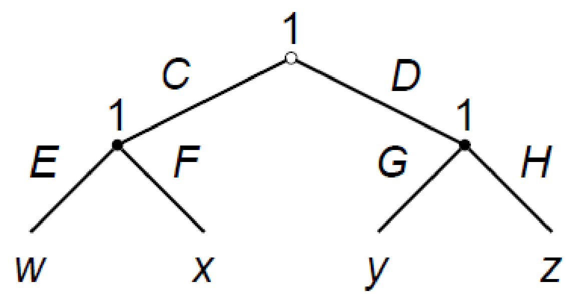
Since we can't backward induct, we need a different general framework

Definition

A strategy profile in an extensive form game[1] satisfies the **one shot deviation principle (OSDP)** if no player can profit by changing a single action at the start of a single subgame given the other players' strategies and the rest of her own

1. with finite horizon or in an infinitely repeated game with discounting

OSDP in a finite extensive form "game"



Suppose $w \geq x \geq y \geq z$

SPE clearly satisfy OSDP, since no change is profitable

Strategy $(C, E(C), G(D))$ satisfies OSDP:

- Cannot improve payoff by switching from E to F if C
- Cannot improve payoff by switching from G to H if D
- Cannot improve payoff by switching from C to D, holding play in the subgames fixed

Other changes $(C, E, G), (D, E, H), (D, F, H)$ not profitable

One-Shot Deviation and SPE

Proposition

Proposition 2. *A strategy profile in a finitely repeated game is a SPE if and only if it satisfies the OSDP*

Proof. “Only if" follows from definition of SPE: if there is a OSD, players are not optimizing at every subgame

For “if", to create a contradiction, posit a profile s that satisfies OSDP but is not SPE

$\exists$ (player i, strategy $\hat{s}_i$, stage τ, history h^τ) such that for i at h^τ, $\hat{s}_i$ yields higher payoff than s_i: $\hat{s}_i(h^\tau \mid s_{-i}) \succ_i s_i(h^\tau \mid s_{-i})$

Let $\hat{\tau}$ be the largest t such that $\hat{s}_i(h^t) \neq s_i(h^t)$ for some h^t; since this is a finitely repeated game, $\hat{\tau}$ is finite

By OSDP $\hat{\tau} > \tau$, since $\hat{s}_i$ must change behavior in more than one place to be better than s_i following h^τ

Proof for the Finite Case (cont'd)

Consider an alternative strategy $\tilde{s}_i$ that agrees with $\hat{s}_i$ at all $t < \hat{\tau}$ and follows s_i from $\hat{\tau}$ on

OSDP implies that s_i is as good a response as $\hat{s}_i$ in every subgame starting at $\hat{\tau}$: $s_i(h^t \mid s_{-i}) \succeq_i \hat{s}_i(h^t \mid s_{-i}) \; \forall h^t : t \geq \hat{\tau}$

This implies that $\tilde{s}_i(h^t \mid s_{-i}) \succeq_i \hat{s}_i(h^t \mid s_{-i}) \; \forall h^t : t \geq \hat{\tau}$

Since $\tilde{s}_i$ equals $\hat{s}_i$ from τ to $\hat{\tau} - 1$, $\tilde{s}_i$ is as good as $\hat{s}_i$ in the subgame starting at τ with history h^τ: $\tilde{s}_i(h^\tau \mid s_{-i}) \succeq_i \hat{s}_i(h^\tau \mid s_{-i})$

- If $\hat{\tau} = \tau + 1$, then $\tilde{s}_i = s_i$ at all but τ where it is same as $\hat{s}_i$; by OSDP $\tilde{s}_i$ cannot improve on s_i: $s_i(h^\tau \mid s_{-i}) \succeq_i \tilde{s}_i(h^\tau \mid s_{-i})$
- Then $\tilde{s}_i$ as good as $\hat{s}_i$ at τ contradicts hypothesis that $\hat{s}_i$ improves on s_i: $s_i(h^\tau \mid s_{-i}) \succeq_i \tilde{s}_i(h^\tau \mid s_{-i}) \succeq_i \hat{s}_i(h^\tau \mid s_{-i})$
- If $\hat{\tau} = \tau + k$, we construct a sequence of k OSD linking $\hat{s}_i$ to s_i, implying $s_i(h^\tau \mid s_{-i}) \succeq_i \hat{s}_i(h^\tau \mid s_{-i})$

A Picture of the Proof Strategy

Stage	τ	$\tau+1$	$\tau+2$	$\ldots$	$\hat{\tau}-2$	$\hat{\tau}-1$	$\mid \hat{\tau}$	$\hat{\tau}+1$	$\ldots$	T
s_i	s_i	s_i	s_i	$\ldots$	s_i	s_i	$\mid s_i$	s_i	$\ldots$	s_i
$\hat{s}_i$	$\hat{s}_i$	$\hat{s}_i$	$\hat{s}_i$	$\ldots$	$\hat{s}_i$	$\hat{s}_i$	$\mid \hat{s}_i$	s_i	$\ldots$	s_i
$\tilde{s}_i$	$\hat{s}_i$	$\hat{s}_i$	$\hat{s}_i$	$\ldots$	$\hat{s}_i$	$\hat{s}_i$	$\mid s_i$	s_i	$\ldots$	s_i
$\tilde{\tilde{s}}_i$	$\hat{s}_i$	$\hat{s}_i$	$\hat{s}_i$	$\ldots$	$\hat{s}_i$	s_i	s_i	s_i	$\ldots$	s_i

Suppose $\hat{t} = \tau + 2$

By OSDP, $\tilde{\tilde{s}}_i$ is as good as $\tilde{s}_i$ in every subgame starting at $\hat{\tau} - 1$; implies $\tilde{\tilde{s}}_i$ as good as $\tilde{s}_i$ as good as $\hat{s}_i$ at τ

$\tilde{\tilde{s}}_i = s_i$ at all but τ (where $\tilde{\tilde{s}}_i = \hat{s}_i$); s_i satisfies OSDP, so $\tilde{\tilde{s}}_i$ cannot improve on s_i; then $\hat{s}_i$ cannot improve on s_i

Same type of argument if $\hat{\tau} = \tau + k$ for any k (it is finite)

OSDP and Infinitely Repeated Games

Extending result to infinitely repeated games a little complicated. Can no longer assert that $\hat{\tau}$ is finite

Events far enough in distant future are essentially irrelevant for today's payoff calculations

At period 1, value of payoff of u from period T on

$$\delta^{T-1}u + \delta^{T}u + \ldots = u\frac{\delta^{T-1}}{1-\delta}$$

As $T \to \infty$ this goes to 0

Doesn't depend on constant per-period payoff, just finite

This property is called **Continuity at Infinity**

OSDP & SPE in Infinitely Repeated Games

Proposition

Proposition 3. *A strategy profile in an infinitely repeated game with $\delta \in (0,1)$ is a SPE if and only if it satisfies OSDP*

Already know that if s satisfies OSDP, cannot be improved by finite sequence of deviations in any subgame. Need to show infinite sequences of deviations do not undo this

Assume, to the contrary, that s satisfies OSDP and is not SPE; $\exists$ (τ, h^τ, i) s.t. i has a better strategy $\overline{s}_i$; let utility improvement be $\epsilon > 0$

By continuity at infinity, $\exists\, \tau'$ such that the strategy s_i' that agrees with $\overline{s}_i$ at all stages before τ' and agrees with s_i at all stages from τ' on must improve on s_i by at least $\epsilon/2$ in the subgame starting at h^τ; this is a finite improvement

Why You Should Love the OSDP

Look for SPE by turning player i's problem into two components: Immediate Payoff and **Continuation Value**

Player i's continuation value from strategy profile $\boldsymbol{s}$ following history $\boldsymbol{h}^t$:

- Denoted $v_i(\boldsymbol{s}(\boldsymbol{h}^t))$
- Average discounted payoff in subgame following $\boldsymbol{h}^t$ if everyone plays according to $\boldsymbol{s}$ following $\boldsymbol{h}^t$

When Does Pl. i Have a Profitable Deviation?

Player i's payoff from playing $s_i(h^t)$ following a history h^t is

$$(1-\delta)u_i(s(h^t)) + \delta v_i(s(h^t, s(h^t)))$$

Player i's payoff from some $s_i' \neq s_i(h^t)$, $s_i' \in A_i(h^t)$ is

$$(1-\delta)u_i(s_i', s_{-i}(h^t)) + \delta v_i(s(h^t, (s_i', s_{-i}(h^t))))$$

Pl. i has no profitable one-shot deviation (and thus no profitable deviation at all) from s at history h^t if

$$\begin{aligned}(1-\delta)u_i(s(h^t)) &+ \delta v_i(s(h^t, s(h^t)))\\ \geq (1-\delta)u_i(s_i', s_{-i}(h^t)) &+ \delta v_i(s(h^t, (s_i', s_{-i}(h^t)))\end{aligned}$$

for all $s_i' \in A_i$

Is Tit-for-Tat a SPE of PD_∞?

Each player plays whatever the other player played last period (Recall condition for NE was $\delta \geq 1/2$)

There are four “classes" of subgames that differ in terms of the payoffs from deviating at them. We must check for one shot deviations at each of them:

1. Histories ending in (C, C)
2. Histories ending in (C, D)
3. Histories ending in (D, C)
4. Histories ending in (D, D)

We will focus on Pl. 1; Pl. 2 is symmetric

One Shot Deviations Following (C, C)

Average discounted payoff (ADP) from adhering[2] is 2.

Following a one shot deviation by Player 1, in which the outcome is (D, C), the outcome alternates between (C, D) and (D, C)

ADP from this one-shot deviation is

$$(1-\delta)3(1+\delta^2+\delta^4+\ldots) = (1-\delta)\left(\frac{3}{1-\delta^2}\right) = \frac{3}{1+\delta}.$$

No profitable deviation if

$$2 \geq \frac{3}{1+\delta} \iff \delta \geq \frac{1}{2}.$$

2. We omit the term δ^{t-1}: it does not matter when deviation takes place

One Shot Deviations Following (C, D)

If adhere, outcome stream alternates between (D, C) and (C, D) yielding

$$(1-\delta)3(1+\delta^2+\delta^4+\ldots) = \frac{3}{1+\delta}$$

If deviate, outcome is (C, C) forever, yielding an average discounted payoff of 2
No profitable deviation if

$$\frac{3}{1+\delta} \geq 2 \iff \delta \leq \frac{1}{2}$$

Exercise

Show that

- No one shot deviations following histories ending in (D, C) requires $\delta \geq 1/2$
- No one shot deviations following histories ending in (D, D) requires $\delta \leq 1/2$

Conclude that both playing Tit-for-Tat is a SPE if and only if $\delta = \frac{1}{2}$

What Payoffs can arise in a SPE of PD_∞?

Proposition

Proposition 4 (SP-Folk Theorem for PD_∞).

- *For any discount factor $\delta \in (0,1)$, the discounted average payoff of each player i in any SPE of PD_∞ is at least $u_i(D,D)$*
- *Let (x_1, x_2) be a feasible pair of payoffs in PD for which $x_i > u_i(D,D)$, for each i. There exists a $\bar{\delta} < 1$ such that if $\delta \geq \bar{\delta}$, then PD_∞ has a SPE in which the average discounted payoff is x_i for each i*
- *For any value of the discount factor, PD_∞ has a SPE in which the average discounted payoff of each player i is $u_i(D,D)$*

Proof of Point 2 of SPE Folk Theorem

Identify sequence of actions $(b^1, b^2, \ldots)$ that generates average discounted payoffs (x_1, x_2)

$$s_i(h^1, \ldots, h^{t-1}) = \begin{cases} b_i^t & \text{if } h^r = b^r \text{ for } r = 1, \ldots, t-1 \\ D & \text{else} \end{cases}$$

Following history $(b^1, \ldots, b^{t-1})$:

- Most profitable possible one-shot deviation yields a payoff of 3 followed by 1 forever
- Not deviating yields x_i
- No deviation if $3(1-\delta) + \delta \leq x_i \iff \delta \geq \frac{3-x_i}{2}$

Following any other history other player plays D forever, so D is obviously best response

The Minmax

For PD_∞, we prove Folk Theorem by threatening worse punishment for deviation that can be forced on a player

Define lowest payoff that can be forced on each player for an arbitrary, infinitely repeated game

For any collection a_{-i} of actions by players other than i, player i's highest possible payoff is $\max_{a_i \in A_i} u_i(a_i, a_{-i})$

To find worst payoff that can be forced on player i in a NE, need the collection a_{-i} that minimizes this maximum:

Definition

Player i's minmax payoff, $\underline{v}_i$, in a strategic game is

$$\underline{v}_i = \min_{a_{-i}} \max_{a_i} u_i(a_i, a_{-i})$$

Nota Bene

Let $\underline{u}_i^*$ the lowest Nash Equilibrium payoff to i. It must be that

$$\underline{v}_i \leq \underline{u}_i^*$$

Proof.

Proof. For some $a_{-i}^* \in \prod_{j \neq i} A_j$, we must have

$$\underline{u}_i^* = \max_{a_i \in A_i} u_i(a_i, a_{-i}^*) \geq \min_{a_{-i} \in \prod_{j \neq i} A_j} \max_{a_i \in A_i} u_i(a_i, a_{-i}) = \underline{v}_i$$

□

The Nash Folk Theorem

Theorem

Let G be a strategic game in which each player has finitely many actions

- For any $\delta \in (0, 1)$, the average discounted payoff of each player in any NE of G_∞ is at least her minmax payoff
- Let w be a feasible payoff profile of G for which each player's payoff exceeds her minmax payoff; $\exists\, \overline{\delta} < 1$ such that if $\delta \geq \overline{\delta}$, then G_∞ has a NE in which the average discounted payoff of each player i is w_i
- If G has a NE in which each player's payoff is her minmax payoff, then for any value of the discount factor, G_∞ has a NE in which the discounted average payoff of each player i is her minmax payoff

Proof of the Nash Folk Theorem

Suppose that profile a is such that each player's average discounted payoff is w_i if a is repeated infinitely

Let p_{-j} be the collection of actions by all other players that holds j to his minmax, and $(p_{-j})_i$ its i^{th} component

Let H^* be the set of terminal histories in which there is at least one period in which the action of exactly one player j differs from a_j

We call a player who is the only one who deviates from a in a given round a *lone deviant*

Proof of the Nash Folk Theorem (cont'd)

Consider the following strategy

$$s_i(h) = \begin{cases} a_i & \text{if } h \notin H^* \\ (p_{-j})_i & \text{if } h \in H^* \text{ and j is the first lone deviant in h} \end{cases}$$

The strategy profile s is a NE for δ sufficiently large:

- If i adheres to s_i, then payoff is w_i
- If i deviates from s_i, payoff from deviating is at most some finite number z; average discounted payoff is $(1-\delta)z + \delta \underline{v}_i$
- j does not deviate if

$$w_i \geq (1-\delta)z + \delta \underline{v}_i \iff \delta \geq \frac{z - w_i}{z - \underline{v}_i} < 1$$

The Problem of Credible Threats

Suppose that the action profile that sustains a player's minmax is not a NE of the stage game

The threat to revert to this profile is not subgame perfect

Our proof of the Nash Folk Theorem is not sufficient to establish the result for SPE

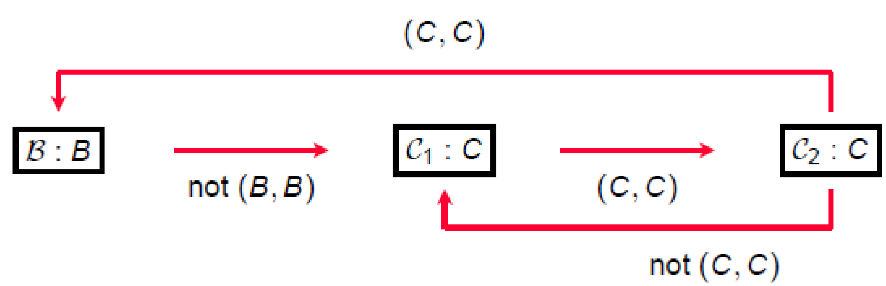
Exercise

		Pl. 2		
		A	B	C
Pl. 1	A	$4, 4$	$3, 0$	$1, 0$
	B	$0, 3$	$2, 2$	$1, 0$
	C	$0, 1$	$0, 1$	$0, 0$

- What is the unique NE?
- What is each player's minmax?
- Try to sustain $(2, 2)$: If Pl. $j \in \{1, 2\}$ doesn't play B, switch to profile yielding j 1 or 0 forever
- Consider the history following a deviation; is switching to this strategy a NE of the subgame?

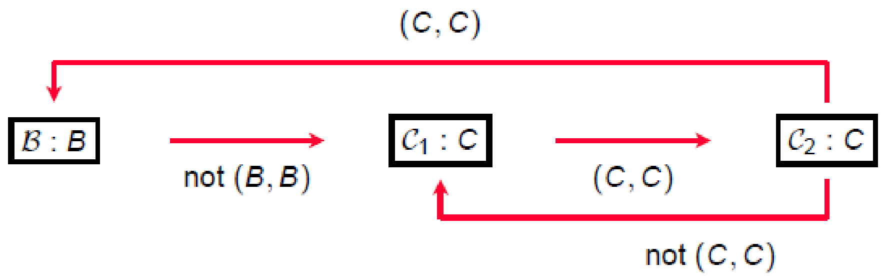
A SPE that Implements (B, B)

		Pl. 2	
	A	B	C
Pl. 1 A	$4, 4$	$3, 0$	$1, 0$
B	$0, 3$	$2, 2$	$1, 0$
C	$0, 1$	$0, 1$	$0, 0$



- Even though not a NE, equilibrium gives both players less than 2 by punishing a failure to punish

One Shot Deviations at $\mathcal{B}$



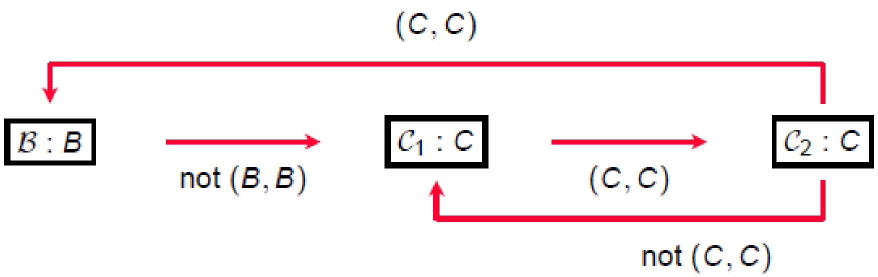
If a player plays B at $\mathcal{B}$, then her payoffs are 2 forever

If deviate to best one-shot deviation, payoffs are $(3, 0, 0)$ followed by 2 forever

No profitable deviation if

$$3 \leq 2(1 + \delta + \delta^2) \iff \delta \geq \frac{1}{2}(\sqrt{3} - 1)$$

One Shot Deviations at $\mathcal{C}_1$



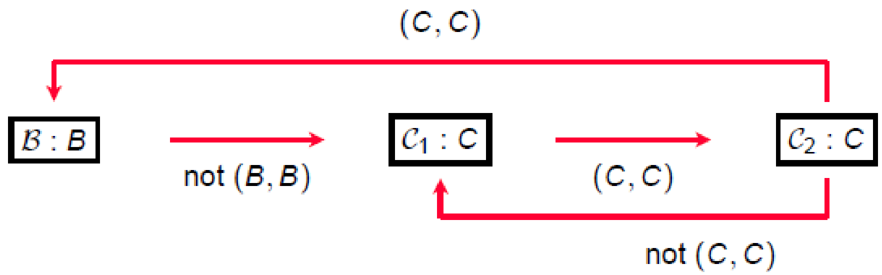
If a player adheres, the payoffs will be $(0, 0, 2)$ and then 2 forever

If a player deviates to best one-shot deviation, her payoffs are $(1, 0, 0)$ and then 2 forever

No profitable deviation if

$$1 \leq 2\delta^2 \iff \delta \geq \sqrt{\frac{1}{2}}$$

One Shot Deviations at $\mathcal{C}_2$



If a player adheres, payoffs will be $(0, 2, 2)$ and then 2 forever

If a player deviates to best one-shot deviation, payoffs will be $(1, 0, 0)$ and then 2 forever

No profitable deviation if

$$1 \leq 2(\delta + \delta^2)$$

which can't bind, since it is weaker than previous condition

Subgame Perfect Folk Theorem

Theorem

Let G be a strategic game in which each player has finitely many actions, and no two players have equivalent payoffs

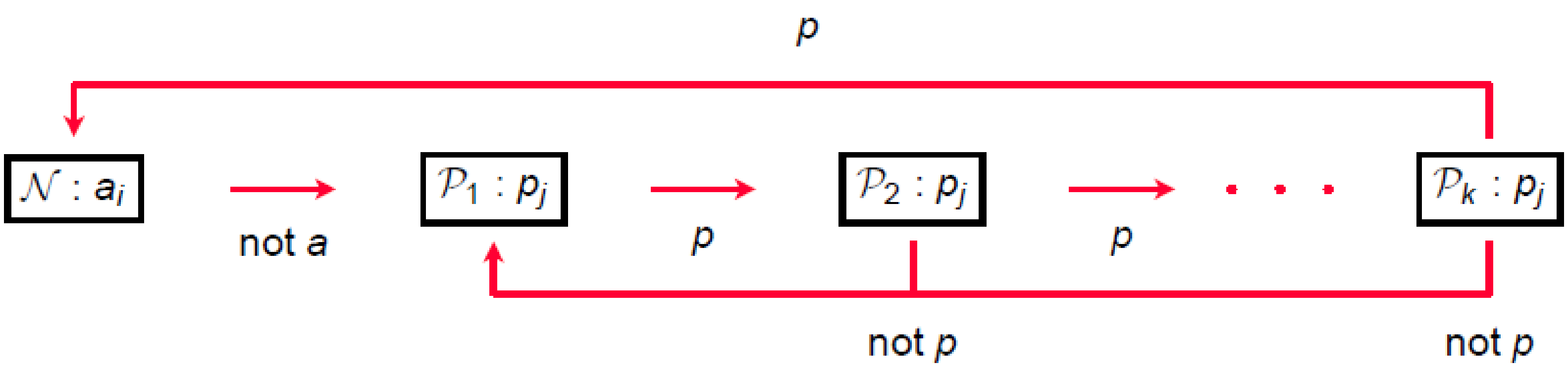
- For any $\delta \in (0, 1)$, the discounted average payoff of each player in any SPE of G_∞ the infinitely repeated game of G is at least her minmax payoff
- Let w be a feasible payoff profile of G for which each player's payoff exceeds her minmax payoff. $\exists\, \overline{\delta} < 1$ such that if $\delta \geq \overline{\delta}$, then G_∞ has a SPE in which the average discounted payoff of each player i is w_i
- If G has a NE in which each player's payoff is her minmax payoff, then for any value of δ, G_∞ has a SPE in which the discounted average payoff of each player i is her minmax payoff

Proof (Two-Player Game)

Let a be an action profile that generates w

Let p_j be an action of player i that holds player j down to her minmax, and $p = (p_2, p_1)$ the strategy where each player holds the other to her minmax

Consider the following strategy, s_i



Proof (Two-Player Game, cont'd)

Claim that strategy profile s generates w in a SPE for a sufficiently large δ for some k

Consider OSD at $\mathcal{N}$

- If deviate get her best payoff from deviation $\overline{u}_i$ this period, $u_i(p)$ for the next k periods, and w for all subsequent periods
- If adhere, gets w_i forever
- Deviation not profitable if

$$w_i(1 + \delta + \ldots + \delta^k) \geq \overline{u}_i + (\delta + \ldots + \delta^k)u_i(p) \iff$$
$$(\delta + \ldots + \delta^k)(w_i - u_i(p)) \geq \overline{u}_i - w_i$$

Since $w_i > u_i(p)$ $\exists$ integer k^* large enough and a discount factor $\delta' < 1$ such that, for $k = k^*$, this inequality is strictly satisfied for $\delta > \delta'$

Proof (Two-Player Game, cont'd)

Consider OSD at any state $\mathcal{P}_l$

- If deviates, payoff is at most her minmax $\underline{v}_i$ in the period of her deviation, $u_i(p)$ for the next k periods, and then w_i forever after
- If adhere, payoff is $u_i(p)$ for next $k - l$ periods and then w_i after that
- Deviation is most tempting at $l = 1$, where it is not profitable if:

$$(1 - \delta + \ldots + \delta^{k-1})u_i(p) + \delta^k w_i \geq \underline{v}_i + (\delta + \ldots + \delta^k)u_i(p)$$
$$\iff \delta^k(w_i - u_i(p)) \geq \underline{v}_i - u_i(p).$$

Since $w_i > \underline{v}_i$, for $k = k^*$ there is a $\delta'' < 1$ such that, for $\delta > \delta''$ this inequality is satisfied

If $\delta > \max\{\delta', \delta''\}$ then s is a SPE sustaining w